25901292 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEGEO201 - CLIMATOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer in 60 Words

Answer any **5** questions

Each question carries **2** marks

1. What is meant by Earth–atmosphere heat system? [K] / [CO2]
2. What is the Coriolis force? [U] / [CO3]
3. Mention one significance of ENSO in climate studies. [K] / [CO4]
4. Define Inter Tropical Convergence Zone (ITCZ). [K] / [CO2]
5. Compare sea breeze and land breeze. [U] / [CO4]
6. Define dew. [K] / [CO5]

[2x5 = 10]

Part B

Short Notes in 250 Words

Answer any **6** questions

Each question carries **8** marks

7. Explain the vertical distribution of temperature in the atmosphere. [U] / [CO1]
8. Describe the distribution of atmospheric pressure on the Earth. [U] / [CO3]

9. Explain different types of anticyclones. [U] / [CO4]

10. Explain the classification of clouds based on height with examples. [U] / [CO4]

11. Explain the reflection of solar radiation and the concept of albedo. [U] / [CO1]

12. Explain precession of the Earth's axis and its impact on seasons and climate. [U] / [CO2]

13. Discuss the polar cell. [U] / [CO2]

14. Discuss the origin of the Indian monsoon briefly. [U] / [CO5]

[8x6 = 48]

Part C

Essay in 400 Words

Answer any **1** questions

Each question carries **12** marks

15. What is the horizontal and vertical distribution of temperature? Explain with suitable examples. [K] / [CO2]

16. Discuss the three main types of rainfall—convectional, orographic, and cyclonic in detail. [U] / [CO5]

[12x1 = 12]